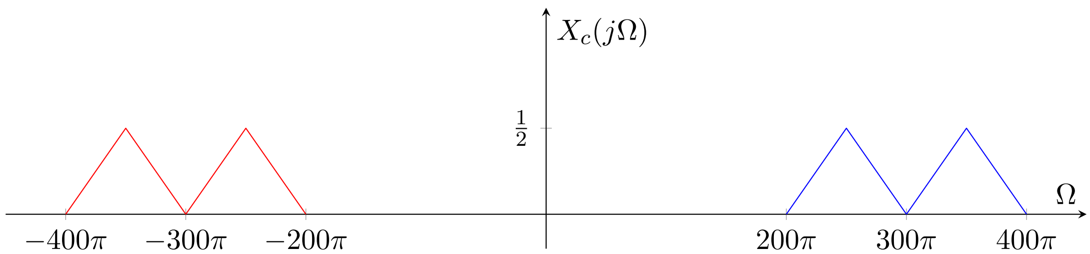
Solutions to Homework #4

Problem 1

(a) The CTFT $X_c(j\Omega)$ is given by

$$X_c(j\Omega) = \frac{1}{2}Z_c(j(\Omega + 300\pi)) + \frac{1}{2}Z_c(j(\Omega - 300\pi))$$

and is plotted as follows.



(b) Since the support of the spectrum is symmetric around the vertical axis, we capture it by two half Nyquist zones. The parameters k and T should satisfy

$$\frac{k\pi}{T} \leq 200\pi \quad \text{and} \quad \frac{(k+1)\pi}{T} \geq 400\pi.$$

Therefore, we need

$$\frac{k\pi}{200\pi} \leq \frac{(k+1)\pi}{400\pi},$$

which is equivalent to

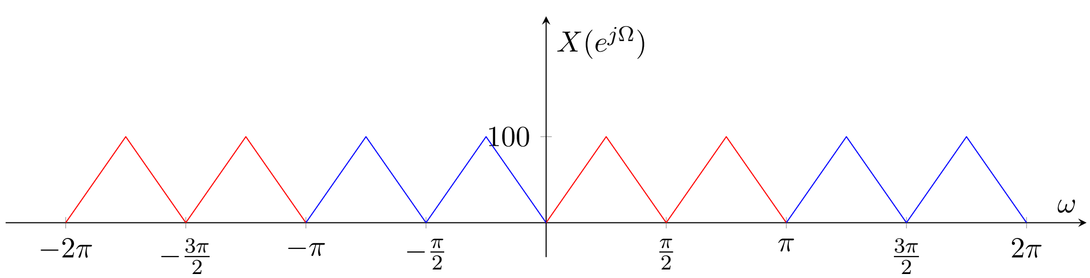
$$2k \leq k + 1.$$

Therefore, the biggest k is 1, which allows $1/T = 200$ Hz.

(c) The DTFT of $x[n]$ is given by

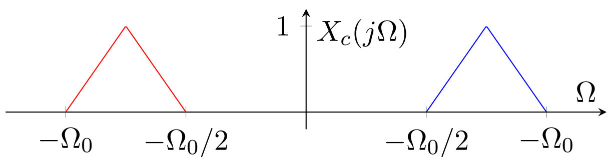
$$X(e^{j\omega}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega - 2\pi k}{T}\right)\right) = 200\sum_{k=-\infty}^{\infty} X_c\left(j200\left(\omega - 2\pi k\right)\right)$$

and plotted as follows.



Problem 2

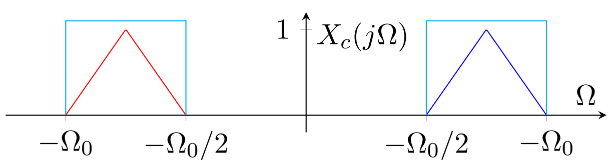
(a) The CTFT $X_c(j\Omega)$ is plotted as follows.



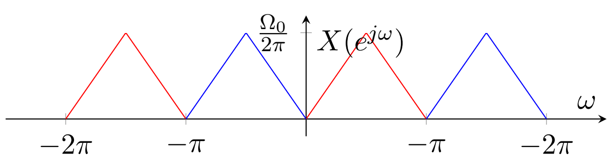
The sampling period T satisfies

$$\frac{\pi}{T} = \frac{\Omega_0}{2}.$$

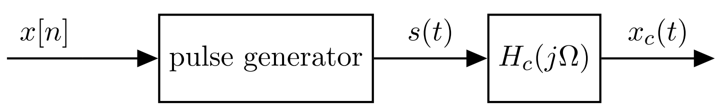
Therefore, the spectrum of the input is captured by the first half Nyquist zones as follows.



After sampling at T, the DTFT $X(e^{j\omega})$ is plotted as follows.



(b) An "idealized" reconstruction scheme is plotted as



where $H_c(j\Omega)$ corresponds to an ideal bandpass filter with the frequency repose

$$H_c(j\Omega) = \begin{cases} T & \text{if } \Omega \in (-\Omega_0, -\Omega_0/2] \cup [\Omega_0/2, \Omega_0) \\ 0 & \text{else.} \end{cases}$$

(c) In part (a), we have considered a bandpass sampling scheme so that the spectrum is captured in the first half Nyquist zones. In fact, to capture the spectrum by any pair of half Nyquist zones, we need to satisfy

$$\frac{\ell\pi}{T} \leq \frac{\Omega_0}{2} \quad \text{and} \quad \frac{(\ell+1)\pi}{T} \geq \Omega_0,$$

which is equivalent to

$$\frac{2\ell\pi}{\Omega_0} \le T \le \frac{(\ell+1)\pi}{\Omega_0}.$$

Here, we will not worry about whether the inequality is strict or not as the spectrum has zero on the boundaries. To have any valid sampling period T satisfy this condition, we need to satisfy $2\ell \le \ell + 1$, therefore, $\ell \le 1$. When $\ell = 1$, it reduces to what we have considered in part (a). On the other hand, if we choose ℓ to be 0, then it reduces to the union of two zeroth half Nyquist zones, which is identical to the zeroth full Nyquist zone. In this case, in order to capture the spectrum by the zeroth Nyquist zone, we need

$$\Omega_0 \le \frac{\pi}{T},$$

which is equivalent to

$$T \le \frac{\pi}{\Omega_0}.$$

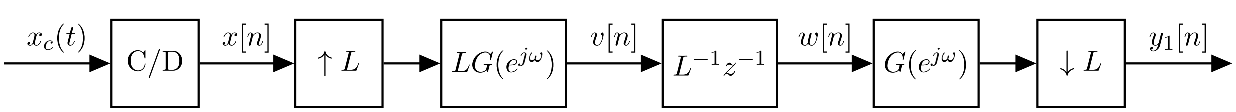
Compared to the bandpass sampling scheme, the sampling period is smaller at least by factor 2. We may conclude that the bandpass sampling is more efficient than the lowpass sampling in this scenario.

Problem 3

Let $G(e^{j\omega})$ denote the ideal lowpass filter given by

$$G(e^{j\omega}) = \begin{cases} 1, & |\omega| \le \pi/L, \\ 0, & \pi/L < |\omega| \le \pi. \end{cases}$$

It trivially satisfies $G(e^{j\omega}) = G(e^{j\omega})G(e^{j\omega})$. Therefore, the system is rewritten as follows.



Note that $v[n]$ is obtained from $x[n]$ by discrete-time interpolation, that is,

$$v[n] = x_c\left(\frac{T}{L}n\right).$$

Then by gain of L^{-1} and delay of 1 sample, we obtain

$$w[n] = \frac{1}{L}\, v[n-1] = \frac{1}{L}\, x_c\left(\frac{T}{L}(n-1)\right) = \frac{1}{L}\, x_c\left(\frac{T}{L}n - \frac{T}{L}\right).$$

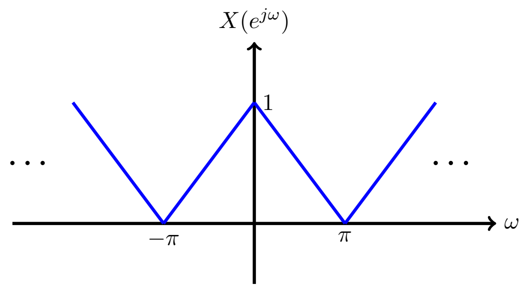
Note that $x_c(Tn/L - T/L)$ is considered as sampled version of $x_c(t - T/L)$ by sampling period T/L. Finally, the last component corresponds to discrete-time decimation by factor L. Therefore, the output is given by

$$y_1[n] = \frac{1}{L}\, x_c\left(Tn - \frac{T}{L}\right).$$

We see that the discrete-time portion of the system (after C/D) implements a factional delay by $1/L$ sample with gain $1/L$.

Problem 4

We are given $X(e^{j\omega})$ as in the following figure.



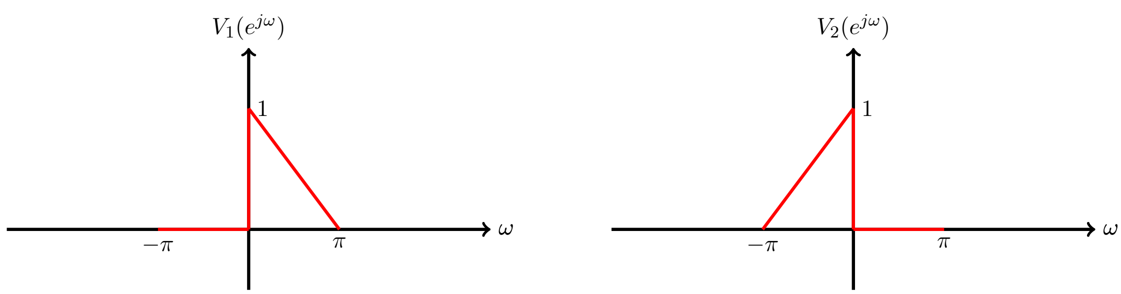
Define

$$H_1(e^{j\omega}) = \begin{cases} 1 & 0 \leq \omega < \pi \\ 0 & -\pi \leq \omega < 0 \end{cases}, \quad H_2(e^{j\omega}) = \begin{cases} 0 & 0 \leq \omega < \pi \\ 1 & -\pi \leq \omega < 0. \end{cases}$$

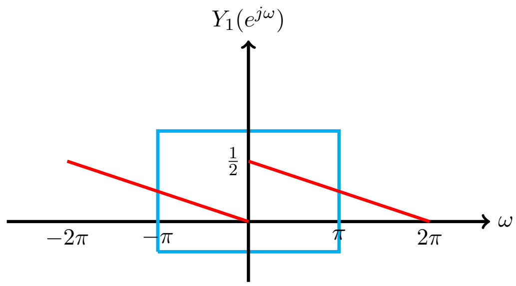
Then the output from $X(e^{j\omega})$ via these filters are given by

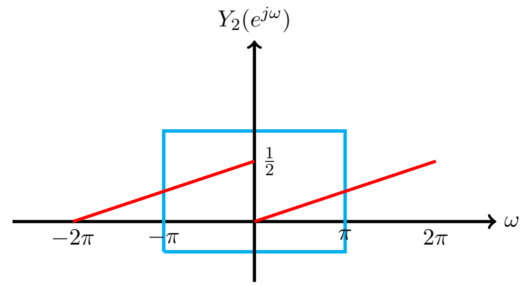
$$V_1(e^{j\omega}) = H_1(e^{j\omega})X(e^{j\omega}), \quad V_2(e^{j\omega}) = H_2(e^{j\omega})X(e^{j\omega}),$$

which are illustrated in the following graphs.



Let $y_1[n]$ and $y_2[n]$ denote the signals respectively from $v_1[n]$ and $v_2[n]$ via downsampling by factor 2, i.e. $y_1[n] = v_1[2n]$ and $y_2[n] = v_2[2n]$. Then, in the frequency domain, it stretches by factor 2 and results in the following graphs.

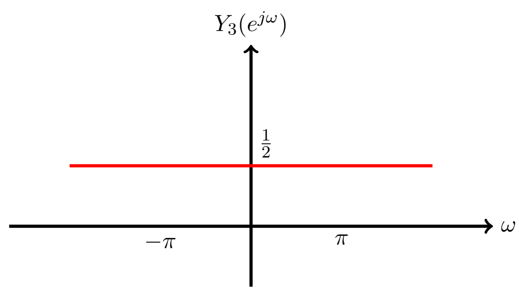




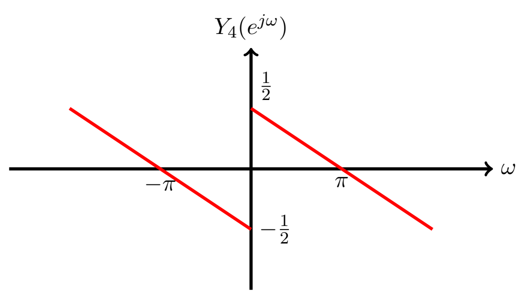
Note that the original filter is synthesized as a linear combination of $H_1(e^{j\omega})$ and $H_2(e^{j\omega})$ given by $H_3(e^{j\omega}) = H_1(e^{j\omega}) + H_2(e^{j\omega})$. Therefore, by the linearity of downsampling, we have,

$$Y_3(e^{j\omega}) = Y_1(e^{j\omega}) + Y_2(e^{j\omega})$$

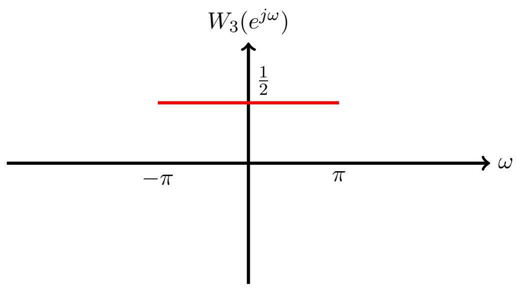
The DTFT of $y_2[n]$ is plotted below.

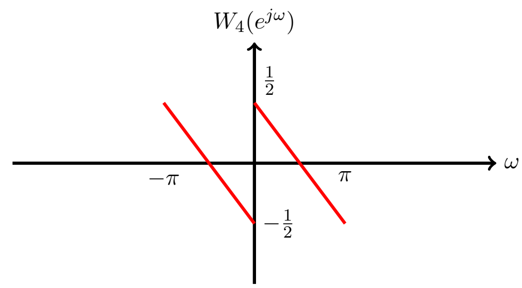


Similarly, since $H_4(e^{j\omega}) = H_1(e^{j\omega}) - H_2(e^{j\omega})$, we obtain that $Y_4(e^{j\omega}) = Y_1(e^{j\omega}) - Y_2(e^{j\omega})$. Then $Y_4(e^{j\omega})$ is plotted as follows:



Let $w_3[n]$ and $w_4[n]$ be the upsampled version of $y_3[n]$ and $y_4[n]$ by factor 2. Then their DTFTs satisfy that $W_3(e^{j\omega}) = Y_3(e^{j2\omega})$ and $W_4(e^{j\omega}) = Y_4(e^{j2\omega})$.

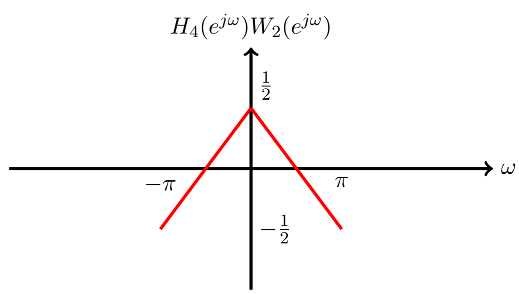




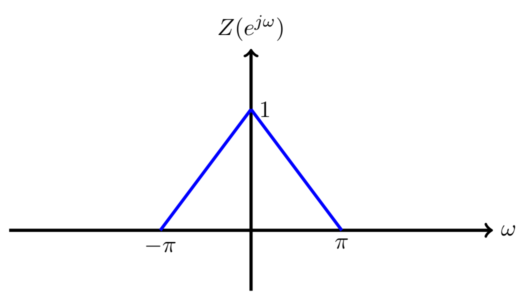
Finally, we reconstruct $z[n]$ by $z[n] = w_3[n] + w_4[n]$. In the frequency domain, we have

$$Z(e^{j\omega}) = W_3(e^{j\omega}) + H_4(e^{j\omega})W_4(e^{j\omega}).$$

Note that $H_4(e^{j\omega})W_4(e^{j\omega})$ is plotted as follows.



Therefore, the final $Z(e^{j\omega})$ is plotted as follows.



Problem 5

(a) The three output signals are written as

$$y_0[n] = x[3n], \quad y_1[n] = x[3n-1], \quad \text{and} \quad y_2[n] = x[3n-2].$$

Let $v_k[n]$ be obtained by upsampling $y_k[n]$ by factor 3 for $k = 0, 1, 2$. Then

$$v_k[n] = \sum_{\langle m \rangle_3 = 0} y_k\left[\frac{m}{3}\right]\delta[n-m] = \sum_{l\in\mathbb{Z}} y_k[l]\delta[n-3l] = \sum_{l\in\mathbb{Z}} x[3l-k]\delta[n-3l].$$

Then

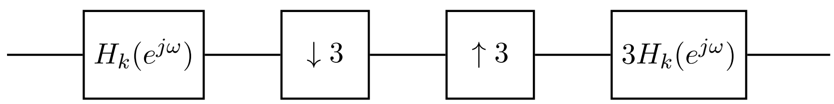
$$v_k[n+k] = \sum_{l \in \mathbb{Z}} x[3l-k]\delta[n+k-3l] = \sum_{l \in \mathbb{Z}} x[3l-k]\delta[n-(3l-k)].$$

Therefore,

$$\sum_{k=0,1,2} v_k[n+k] = \sum_{k=0,1,2} \sum_{l \in \mathbb{Z}} x[3l-k]\delta[n-(3l-k)] = \sum_{m \in \mathbb{Z}} x[m]\delta[n-m] = x[n].$$

We have shown that we can exactly reconstruct $x[n]$ from $y_0[n]$, $y_1[n]$, and $y_2[n]$.

(b) Note that $H_k(e^{j\omega})$ is supported on the union of two intervals of length $\frac{\pi}{3}$ so that the aliased copies from downsampling and upsampling do not overlap. Therefore, the system in the following block diagram



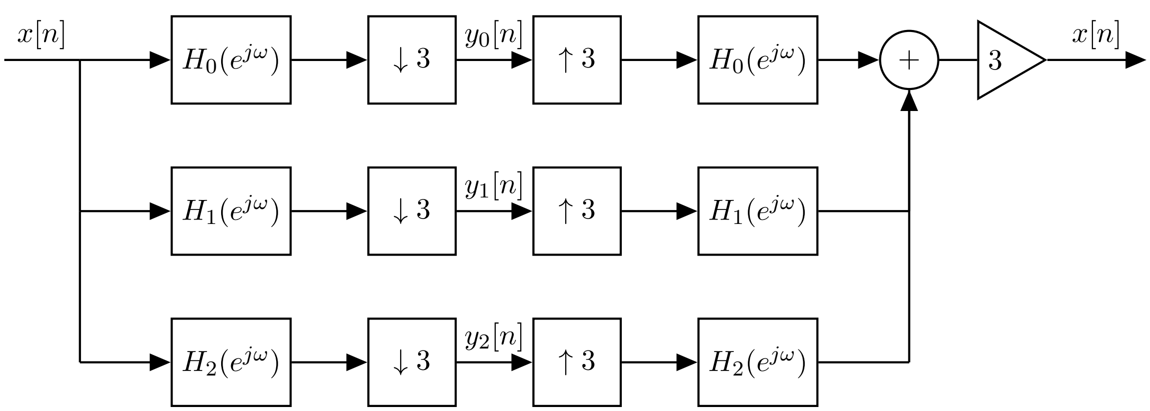
is equivalent to the LTI system for $H_k(e^{j\omega})$. Furthermore, we have

$$H_0(e^{j\omega}) + H_1(e^{j\omega}) + H_2(e^{j\omega}) = 1.$$

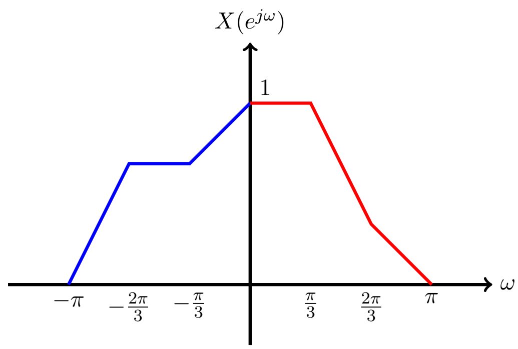
Then it follows that

$$\begin{aligned}
&3H_0(e^{j\omega})Y_0(e^{j3\omega}) + 3H_1(e^{j\omega})Y_1(e^{j3\omega}) + 3H_2(e^{j\omega})Y_2(e^{j3\omega}) \\
&= H_0(e^{j\omega})X(e^{j\omega}) + H_1(e^{j\omega})X(e^{j\omega}) + H_2(e^{j\omega})X(e^{j\omega}) \\
&= \left(H_0(e^{j\omega}) + H_1(e^{j\omega}) + H_2(e^{j\omega})\right) X(e^{j\omega}) \\
&= X(e^{j\omega}).
\end{aligned}$$

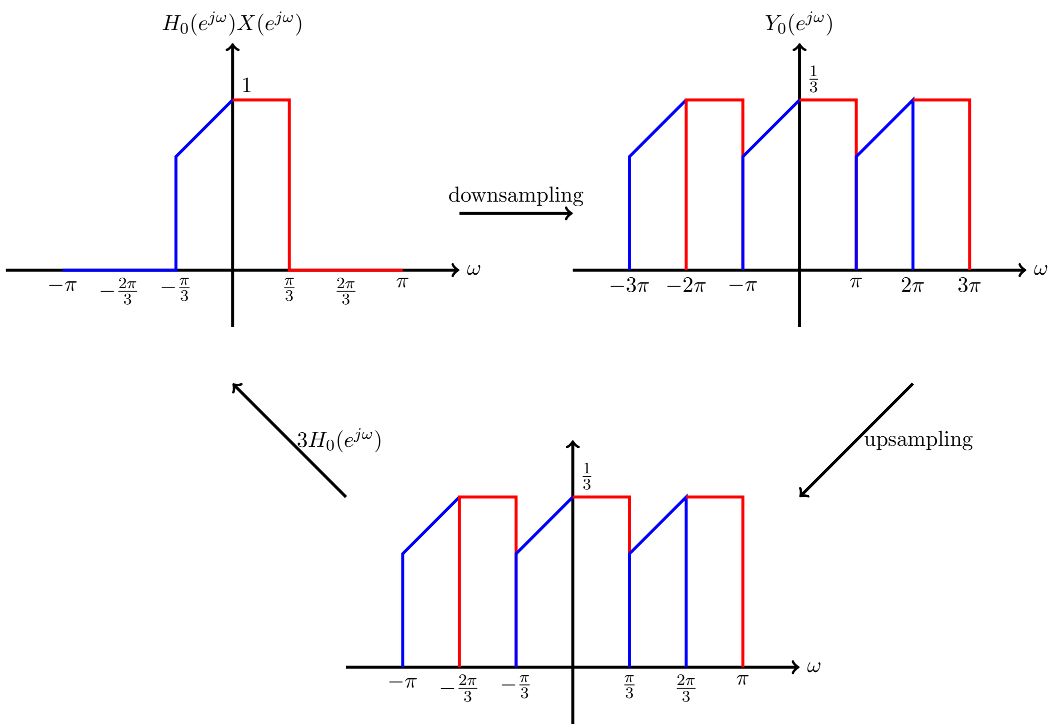
This is illustrated as the following block diagram.

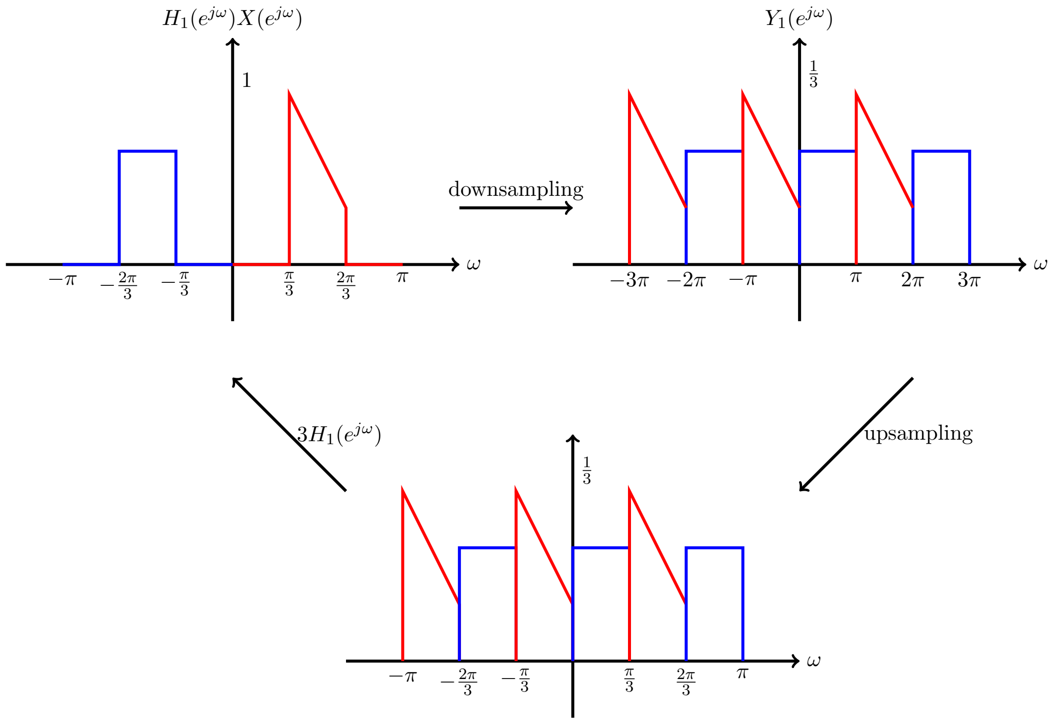


Let us illustrate what happens in the frequency domain through a concrete example. Suppose that $X(e^{j\omega})$ is shaped as in the following graph. The shape of $X(e^{j\omega})$ was chosen arbitrarily.



Then the DTFTs vary according to the filtering, downsampling, and upsampling as shown below.





$H_1(e^{j\omega})X(e^{j\omega})$
1
ω
$-\pi$
$-\frac{2\pi}{3}$
$-\frac{\pi}{3}$
$\frac{\pi}{3}$
$\frac{2\pi}{3}$
π
downsampling
$Y_1(e^{j\omega})$
$\frac{1}{3}$
ω
-3π
-2π
$-\pi$
π
2π
3π
upsampling
$3H_1(e^{j\omega})$
$\frac{1}{3}$
ω
$-\pi$
$-\frac{2\pi}{3}$
$-\frac{\pi}{3}$
$\frac{\pi}{3}$
$\frac{2\pi}{3}$
π